

Week 37: Long-Run Behavior of Markov Chains (Part 2)

Recap from last week

Definition 0.1. (*Limiting distribution*) Consider a chain $\{X_n\}$. $\pi = (\pi_0, \pi_1, \dots)^\top$ is the *limiting distribution* if:

1. The limits $\pi_j = \lim_{n \rightarrow \infty} P_{i,j}^{(n)}$ exist for all i and j .
2. $\sum_{j=0}^{\infty} \pi_j = 1$.

Definition 0.2. (*Regular Markov chain*) Consider a chain $\{X_n\}$ with finite state space $\{0, 1, \dots, N\}$ and transition probability matrix $\mathbf{P}$. If there exists a positive integer k such that all elements of $\mathbf{P}^k$ are strictly positive, we call $\mathbf{P}$ and $\{X_n\}$ *regular*.

I maybe confused you a bit with stationary vs. limiting distributions. We'll return to this with more "rigour"!

Week 37

Definition 1.3. (*Communication*) Let $\{X_n : n = 0, 1, \dots\}$ be a Markov chain with transition probability matrix $\mathbf{P}$.

- State j is *accessible* from state i if there exists an integer $n \geq 0$ such that $P_{i,j}^{(n)} > 0$.
- If states i and j are accessible from each other, they *communicate*. Write as $i \sim j$.

Theorem 1.4. (*Communication is an equivalence relation*)

- *Reflexivity*: $i \sim i$ for every state i .
- *Symmetry*: if $i \sim j$, then $j \sim i$.
- *Transitivity*: if $i \sim j$ and $j \sim k$, then $i \sim k$.

Relation partitions the state space into *equivalence classes* of communicating states.

Definition 1.5. (*Irreducibility*) A Markov chain is *irreducible* if it has exactly one communication class, meaning that all states communicate. Otherwise, the chain is *reducible*.

Problem

Consider two Markov chains with transition probability matrices

$$\mathbf{A} = \begin{bmatrix} 1/3 & 1/3 & 1/3 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

For each chain, determine how many equivalence classes it has and whether it is irreducible or reducible.

Definition 1.6. (*Periodicity*) The *period* of state i , denoted by $d(i)$, is

$$d(i) = \gcd\left\{n \geq 1 : P_{i,i}^{(n)} > 0\right\}.$$

If $P_{i,i}^{(n)} = 0$ for every $n \geq 1$, we define $d(i) = 0$.

Theorem 1.7. If $i \sim j$, then $d(i) = d(j)$.

Notes:

- A state with $d(i) = 1$ is called *aperiodic*.
- All states in an irreducible chain have the same period, so we call the chain *periodic* or *aperiodic* accordingly.
- If state i is aperiodic, there exists an integer m_i such that $P_{i,i}^{(n)} > 0$ for every $n \geq m_i$ (used in a proof later).

Problem

Consider the Markov chains specified by the transition probability matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & \cdots & 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1/2 & 0 & 1/2 & 0 \end{bmatrix}.$$

Assume $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$.

For each chain do the following:

- Is the Markov chain reducible or irreducible? If it is reducible, specify its equivalence classes.
- Calculate the period of each state.

Definition 1.8. (*First-return probabilities*)

$$f_{i,i}^{(n)} = P(X_n = i, X_\nu \neq i \text{ for } \nu = 1, 2, \dots, n-1 \mid X_0 = i), \quad n \geq 1,$$

and we set $f_{i,i}^{(0)} = 0$. The probability of ever returning to state i is $f_{i,i} = \sum_{n=1}^{\infty} f_{i,i}^{(n)}$.

Definition 1.9. (*Recurrent and transient states*) State i is *recurrent* if $f_{i,i} = 1$. It is *transient* if $f_{i,i} < 1$.

Theorem 1.10. *Recurrence and transience are class properties. If $i \sim j$ and i is recurrent, then j is recurrent.*

Theorem 1.11. *A state i is recurrent if and only if $\sum_{n=1}^{\infty} P_{i,i}^{(n)} = \infty$.*

Problem

A two-state Markov chain has the transition probability matrix

$$\mathbf{P} = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}.$$

Determine the first-return distribution

$$f_{0,0}^{(n)} = P(X_n = 0, X_\nu \neq 0 \text{ for } \nu = 1, 2, \dots, n-1 \mid X_0 = 0), \quad n \geq 1,$$

Problem: Proving a theorem

Let $\{X_n : n = 0, 1, \dots\}$ be a Markov chain with finite state space $\{0, 1, \dots, N\}$. Show that if the Markov chain is aperiodic and irreducible, then it is regular and recurrent.

(Recall that a Markov chain is regular when there exists a positive integer k such that all elements of $\mathbf{P}^k$ are strictly positive.)

Definition 1.12. (*Mean recurrence time*) The mean recurrence time of state i is

$$m_i = \sum_{n=1}^{\infty} n f_{i,i}^{(n)},$$

and represents the expected time between successive visits to state i .

Definition 1.13. (*Positive and null recurrence*) A recurrent state i is positive recurrent if $m_i < \infty$, null recurrent if $m_i = \infty$.

Positive recurrence and null recurrence are also class properties.

Definition 1.14. (*Stationary distribution*) A probability vector $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots)^\top$ is a stationary distribution if

$$\sum_{i=0}^{\infty} \pi_i = 1 \quad \text{and} \quad \pi_j = \sum_{i=0}^{\infty} \pi_i P_{i,j}, \quad j = 0, 1, \dots$$

For finite state spaces, $\boldsymbol{\pi} = \mathbf{P}^\top \boldsymbol{\pi}$ and $\mathbf{1}^\top \boldsymbol{\pi} = 1$.

Theorem 1.15. (*Existence and uniqueness of stationary distributions*) An irreducible Markov chain has a stationary distribution if and only if it is positive recurrent.

In that case, the stationary distribution is unique and $\pi_i = 1/m_i$.

Theorem 1.16. *(When the stationary distribution is limiting) An irreducible, positive recurrent, and aperiodic Markov chain has a (unique) limiting distribution.*

It is equal to the (unique) stationary distribution

$$\lim_{n \rightarrow \infty} P_{j,i}^{(n)} = \pi_i = \frac{1}{m_i}$$

for all states i, j .

Corollary 1.17. *(Finite irreducible chains) Every finite irreducible Markov chain is positive recurrent, and therefore has a unique stationary distribution. If it is also aperiodic, this stationary distribution is also the limiting distribution.*

In general (both finite and infinite):

Irreducible + Positive recurrent $\implies$ Unique stationary distribution

Irreducible + Positive recurrent + Aperiodic $\implies$ Unique limiting/stationary distribution

Finite Markov chains: Irreducible $\implies$ Positive recurrent

Problem: Exam 2023

Consider the Markov chain $\{X_n : n = 0, 1, \dots\}$ with state space $\{0, 1, 2\}$, and transition probability matrix

$$\mathbf{P} = \begin{bmatrix} \alpha & 1 - \alpha & 0 \\ 0 & \beta & 1 - \beta \\ 1 - \gamma & 0 & \gamma \end{bmatrix}$$

for some known $\alpha, \beta, \gamma \in [0, 1]$.

- What conditions are necessary for the Markov chain to be irreducible?
- Assume the irreducibility condition holds:
 - What are the equivalence classes? What are the periodicities of each state, and under what conditions?
 - Is the Markov chain positive recurrent? Does the chain have a stationary distribution? If so, is it a limiting distribution?

Problem

We throw a fair coin repeatedly, and denote each outcome by H (heads) or T (tails). Let n denote the n -th throw in a sequence, and define a stochastic process $\{X_n : n = 0, 1, \dots\}$ by

$X_n =$ The number of consecutive throws of H, including throw n , $n = 0, 1, \dots$

Do the following:

- Explain why this is a Markov process. Specify the state space, and find the transition probabilities.
- Is this Markov chain irreducible?
- For each state, calculate its period.
- Show that the states are recurrent.
- Calculate the long-run proportion of time that the sequence of throws ends in three or more heads.